

EXOPLANETS

The image is a full-page background with a deep space theme. It features a large, dark, cratered planet in the upper left quadrant, a smaller similar planet in the lower right, and a bright, glowing nebula in the center. The word 'EXOPLANETS' is written in a bold, white, stylized font across the middle. The background is a dark blue and purple space filled with stars and a nebula.

What is an exoplanet?

- Any planet found outside our solar system
- “extrasolar planet”

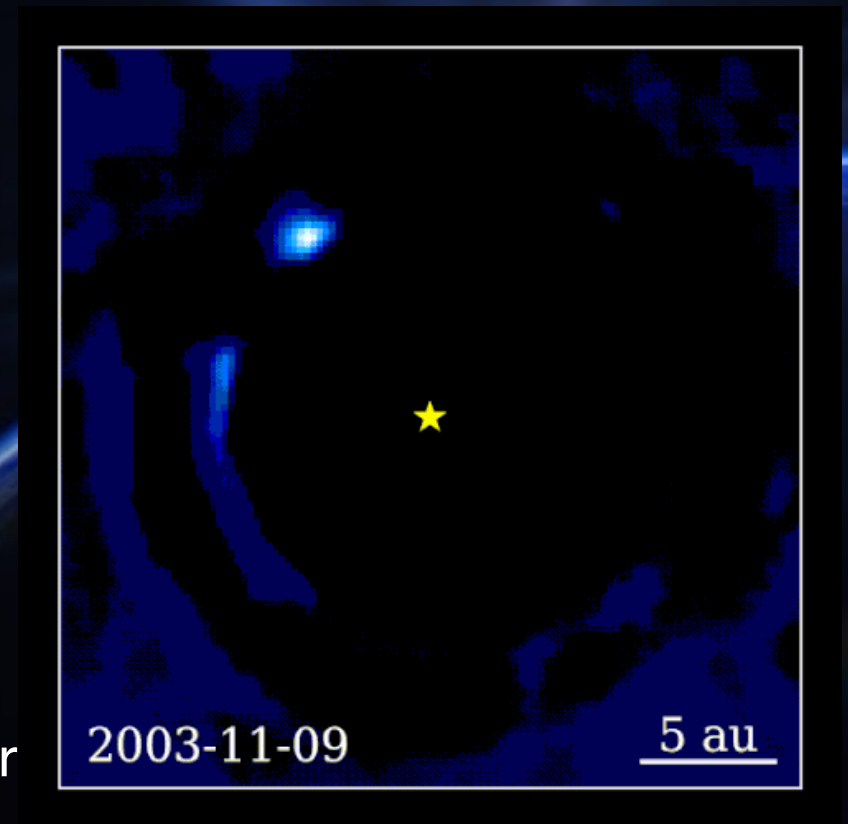
When were exoplanets first discovered?

- First confirmed discovery was in 1992
- 3 planets (named Draugr, Poltergeist, and Phobos) are “hot Jupiters” orbiting a pulsar– not what astronomers expected
- The first exoplanets orbiting a main sequence star were discovered in 1995. Scientists were looking for solar systems similar to ours.
- Many discoveries were claimed before 1992, but none could be confirmed

Why are exoplanets hard to detect?

- Usually smaller than stars
- Do not emit light or radio waves
- Reflected light is dim compared to stars
- Mass is low compared to stars

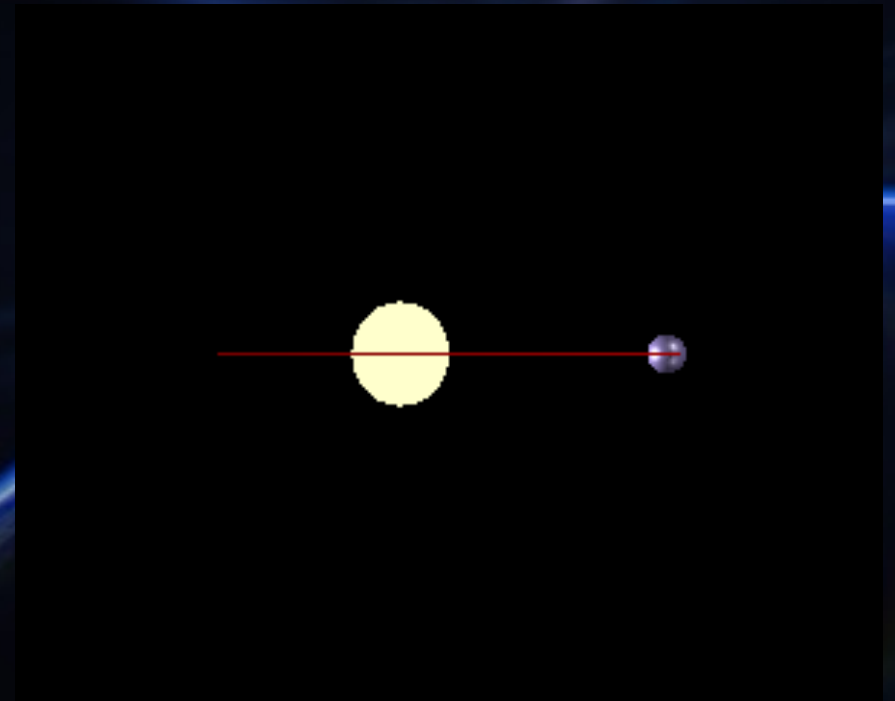
This is a direct image Beta Pictoris b.
Direct imaging of exoplanets is very difficult because the planet is very dim compared to its star.
Direct imaging works best for planets that are very large and very far away from a star.



How do we detect exoplanets?

Transit Method

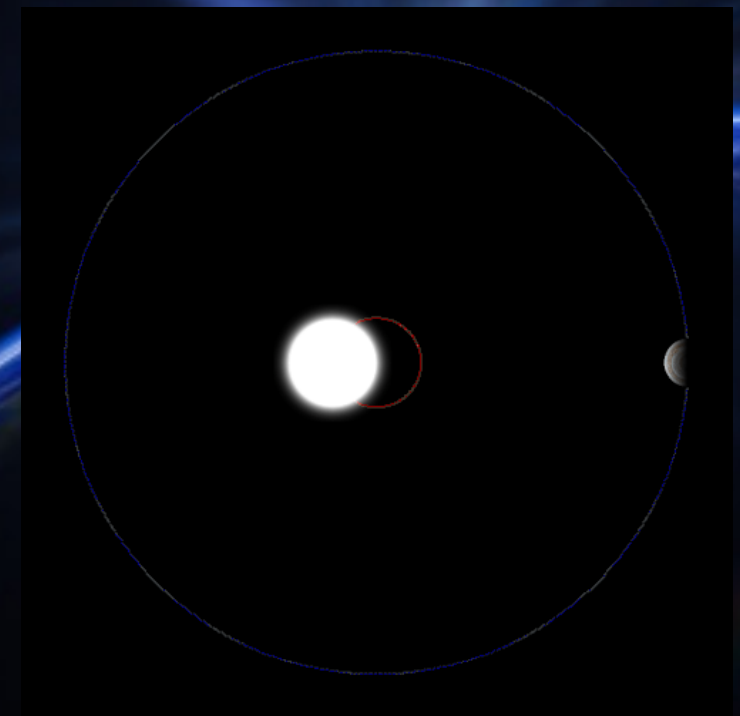
- The star's light dims as the planet passes in front of it.
- Depends on the location of the observer
- Works best for large planets close to the star
- Planets with long orbits take longer to see
- Many stars can be studied at the same time using this method



How do we detect exoplanets?

Radial Velocity or Doppler Method

- The planet's gravity causes the star to revolve around the center of mass of the planetary system.
- The star's "wobble" toward or away from Earth can be precisely measured
- Works best for large planets close to the star
- Earth-mass planets can only be detected this way if they are very close to the star



How do we detect exoplanets?

Gravitational Microlensing

- A star's gravity magnifies the appearance of distant stars like a lens
- A planet's pull on the star causes small changes to the magnification, which can be measured
- Best method for finding planets 1-10 AU from Sun-like stars



Pulsar Timing

- A pulsar is a star that has exploded in a supernova and emits regular bursts of radio waves
- A planet's gravitational pull disrupts the timing of the radio waves
- The first exoplanets were discovered this way
- Only works for pulsars

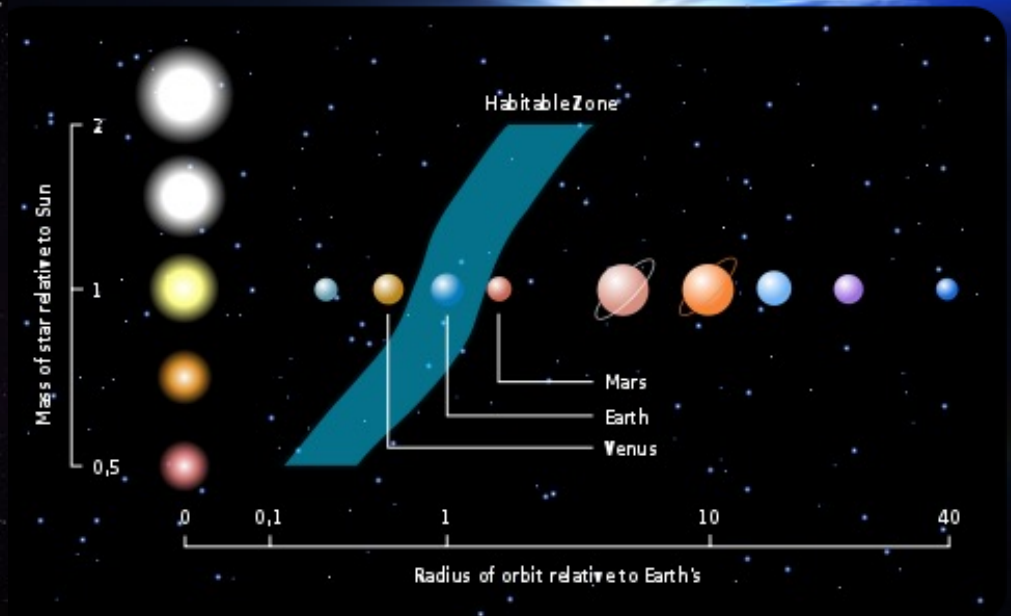
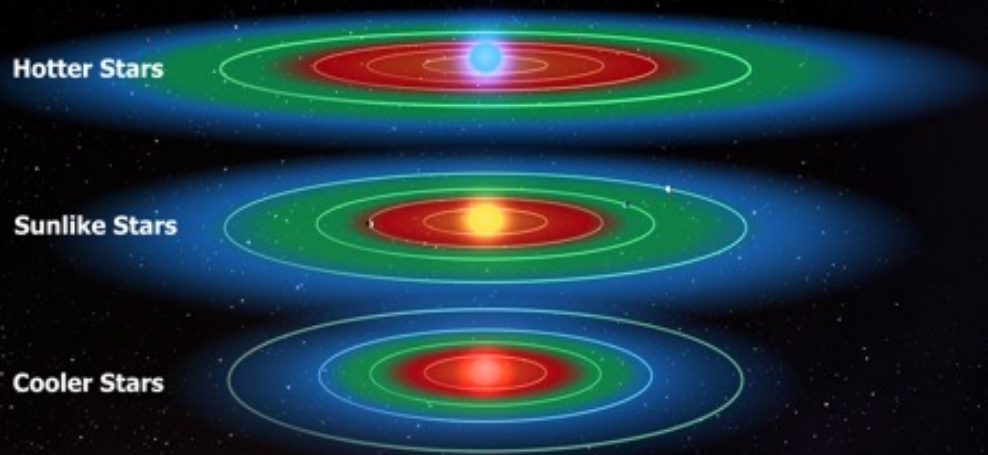


Is there life on exoplanets?

No life has yet been detected on any other planet, but astrobiologists are looking!

One starting point is the circumstellar habitable zone (CHZ)

This is the distance from a star where water can stay liquid on the surface



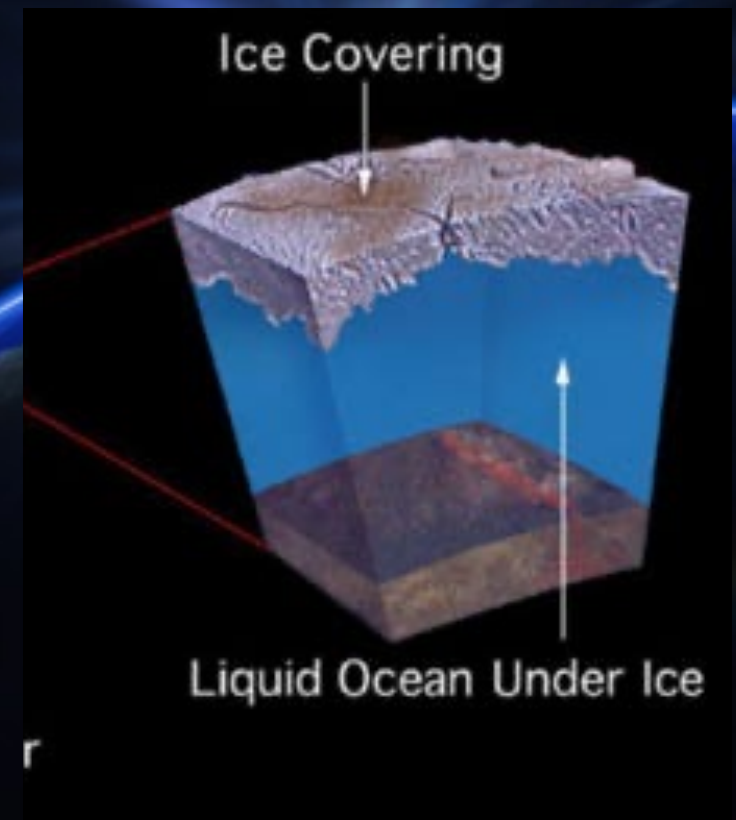
Is there life on exoplanets?

Astrobiologists are also looking for life outside the CHZ

Many planets like Pluto have liquid oceans beneath surface ice

These oceans can be kept warm by:

1. tidal heating- when the planet is squished and expanded by gravity from a star, moon, or another planet, the friction produces heat
2. radioactive decay- radioactive material from the planet's formation release heat over time
3. Slow cooling- planets may retain heat from formation for a long time



What types of exoplanets have we found?

- As of May 8th, 2025, there are 5,893 confirmed exoplanets
- 4,398 stars are confirmed to have planets
- 987 stars are confirmed to have more than one planet
- Astronomers now think there may be more planets in our galaxy than stars
- Astronomers group planets by their size and average temperature, but also the type of star they orbit, and other features
- **Rogue planets** do not orbit a star and are thought to greatly outnumber star-bound exoplanets. These may form on their own or be ejected when planetary systems like our solar system form

What types of exoplanets have we found?

Exoplanet Types



Gas Giants

The size of Saturn or Jupiter, or much larger. They include "hot Jupiters"- scorching planets in close orbits around their stars.



Neptune-Like

Similar in size to our own Neptune and Uranus, with hydrogen or helium-dominated atmospheres. "Mini-Neptunes," not found in our solar system, are smaller than Neptune but larger than Earth.



Terrestrial

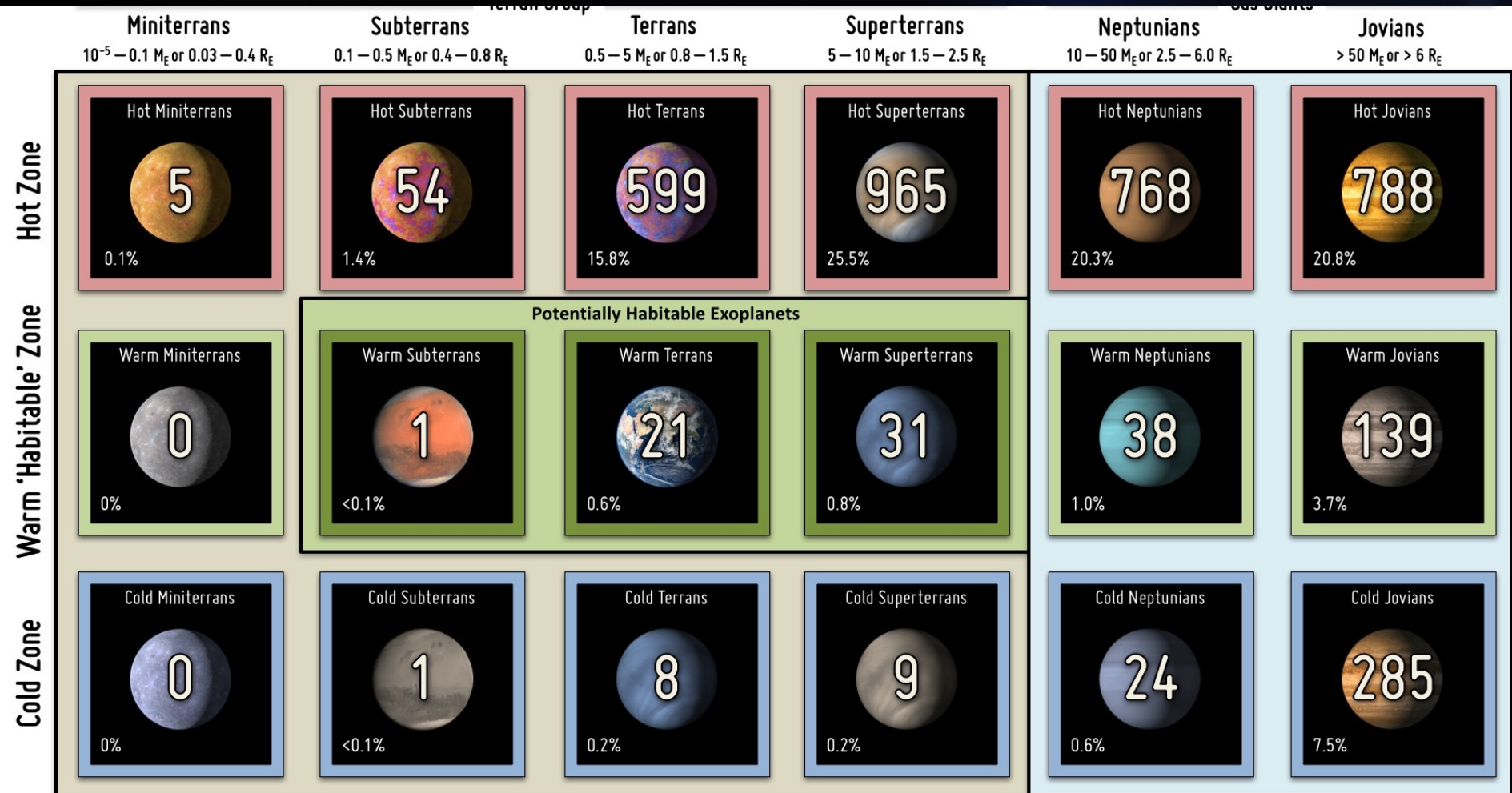
Earth-sized or smaller, mostly made of rock and metal. Some could possess oceans or atmospheres and perhaps other signs of habitability.



Super-Earth

Typically "terrestrial," or rocky, and more massive than Earth but lighter than Neptune. They might or might not have atmospheres.

What types of exoplanets have we found?



M_E = Earth Mass, R_E = Earth Radius

CREDIT: PHL @ UPR Arcibo (phl.upr.edu) Nov 2017

